

P Akshay Thomas

AI ENGINEER — AGENTIC SYSTEMS, RAG & LLM PLATFORMS

+91-7993129238 | akshaythomas.p@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI Engineer with 3.5+ years building production LLM systems at enterprise scale — including an agentic Multi-Agent Platform, the company's sole company-wide RAG system spanning four business domains, MCP servers giving senior executives direct natural-language access to Salesforce and Atlassian through Claude, and several other production AI applications built and deployed end-to-end across the org.

EXPERIENCE

RINGCENTRAL INNOVATION INDIA

Sept 2022 – Present

Software Development Engineer 2 — AI Agent & LLM Platforms (Integrations & Automations Team)

- Architected a **Multi-Agent Platform (LangGraph, FastAPI)** with a Dynamic Schema Registry in **PostgreSQL** that decouples agent definitions from the codebase, enabling new agents to be registered and deployed without code changes or redeployment; instrumented the routing layer with **MLflow** for run tracking and observability; shipped real-time token streaming via **SSE** and persisted conversation history in **BigQuery** for multi-turn context retrieval and compliance logging.
- Built the company's **sole production RAG system** serving as the authoritative company-wide knowledge source across **four business domains** (HR, IT, OTC, RC Support) on **Pinecone**, indexing **100k+ enterprise vectors** with hybrid retrieval (dense: **gemini-embedding-001**; sparse: **pinecone-sparse-english-v0**); designed heading-aware chunking with breadcrumb prefixes, domain namespace isolation, and metadata-driven region filtering with global fallback, plus incremental re-ingestion to skip unchanged content; built a **golden-dataset evaluation harness** achieving **85–95% accuracy**, tracked via **MLflow** — deployed on **Google Cloud Run** with streaming answers.
- Engineered production **MCP servers for Claude** giving **senior executives and business leaders** direct natural-language access to **Atlassian** (Jira, Confluence) and **Salesforce**, letting non-technical leadership query and act on live enterprise systems directly; built a custom **OAuth 2.1** authentication layer (Google-backed) with per-user credential management via a **Firestore**-persisted PAT registry and **JWT Bearer** auth for Salesforce; deployed both servers on **Google Cloud Run** over **Streamable HTTP**.
- Eliminated hallucination risk in numeric outputs by engineering a **Document Analytics agent (Google Gemini 2.5 via Vertex AI)** with a Local Execution Architecture to process datasets beyond context limits, plus a regex-based query sanitization layer that validates LLM-generated **Pandas** code before execution — ensuring calculations run on verified code rather than LLM-guessed numbers.
- Shipped a conversational **NetSuite AI Agent**, adopted by the **full procurement team** for daily natural-language querying of procurement data, using a hybrid routing architecture (Gemini Flash-lite for intent routing, Flash for response generation) backed by a **PostgreSQL** session-management layer that persists multi-turn conversation state across complex context switches.

Software Development Engineer 1 (Integrations & Automations Team)

- Automated enterprise procurement/sales workflows (**RingCentral ↔ Coupa/Salesforce**) via **AWS Serverless (Lambda, DynamoDB, SQS)** in **TypeScript**, cutting executive approval times by **60%**; also built **Ebonding** workflows and REST APIs on **Dell Boomi** for B2B partners, plus separate internal integrations syncing **Workday, NetSuite, Salesforce, Jira, Freshservice, and DWH**.

PROJECTS

The Interval — Multi-Agent World Simulation

[GitHub](#)

- Built a persistent multi-agent simulation where AI agents (portraying historical figures) converse over the **A2A protocol** and can only mutate world state through validated **MCP tools**; instrumented every LLM call with **Langfuse** tracing, added a security-focused admission gate (schema & injection-refusal checks for new agents), and backed it with a **118-test** suite — running entirely on free-tier infrastructure.

SKILLS

LLM & Agentic Systems: LangGraph, LangChain, Multi-Agent Systems, Agentic AI, RAG, MCP (Model Context Protocol), Prompt Engineering, Vertex AI, Gemini

Vector Search & Retrieval: Pinecone, Hybrid Dense/Sparse Retrieval, Embeddings, Chunking Strategies

Evaluation & Observability: Golden-Dataset Evals, MLflow, Langfuse, Prompt/Trace Monitoring

Languages & Backend: Python (FastAPI, Flask, Pandas), TypeScript (Node.js), C++, SQL (MySQL, PostgreSQL)

Cloud & DevOps: GCP (Vertex AI, Cloud Run, BigQuery, Cloud Build, Firestore), AWS (Lambda, DynamoDB, SQS), Docker, Git, CI/CD

Architecture & Integration: REST APIs, Server-Sent Events (SSE), OAuth 2.1, Serverless, Message Queues, Dell Boomi (iPaaS)

AI-Assisted Frontend: React, TypeScript (Production-deployed UIs using AI coding tools)

EDUCATION

Indian Institute of Technology Madras

Jul 2018 - May 2022

B.Tech in Computer Science & Engineering

AWARDS & ACHIEVEMENTS

- Awarded **Ace of the Quarter** (Q1 & Q3, 2024) and mentored **4 engineering interns** through to full-time conversion.
- Secured All India Rank **186** in **JEE Advanced 2018** (top 0.1 percentile nationally).